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SYSTEMS AND METHODS FOR ASSESSING OPERATIONAL INTEGRITY OF ULTRASONIC TRANSDUCERS

Abstract

A system for assessing operational integrity of an ultrasonic transducer includes a metrological tool including a rod having test holes. The ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the rod. A method for assessing operational integrity of an ultrasonic transducer includes coupling the ultrasonic transducer to a metrological tool including a rod having test holes, and emitting, by the ultrasonic transducer, ultrasonic energy into the rod.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] Examples of the present disclosure generally relate to systems and methods for assessing and verifying operational integrity of ultrasonic transducers, such as can be used to non-destructively inspect various structural components of an aircraft.

BACKGROUND OF THE DISCLOSURE

[0002] Ultrasonic transducers can be used to inspect various structural components. For example, during a manufacturing process of a commercial aircraft, ultrasonic transducers can be used to non-destructively inspect various structures, such as stringers, spars, frames, and the like of various portions of the aircraft.

[0003] Typically, ultrasonic transducers having ultrasonic linear arrays are custom manufactured for non-destructive inspection of particular components. Performance of an ultrasonic transducer varies based on a particular component being inspected.

[0004] In general, operational performance of an ultrasonic transducer used for non-destructive inspection is unknown until the ultrasonic transducer is actually used during an inspection process. However, the unknown condition can impact production by creating delays in system qualification.

SUMMARY OF THE DISCLOSURE

[0005] A need exists for a system and a method for verifying operational integrity of an ultrasonic transducer, such as used for non-destructive inspection, before being shipped to an end user.

Because ultrasonic transducers used for non-destructive inspection are typically custom manufactured, there can be a long lead time before delivery to an end user. As such, there is a need to verify operational verification of the ultrasonic transducer.

[0006] With those needs in mind, certain examples of the present disclosure provide a system for assessing operational integrity of an ultrasonic transducer. The system includes a metrological tool including a rod having test holes. The ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the rod.

[0007] In at least one example, a control unit is in communication with the ultrasonic transducer. The control unit is configured to generate a sonogram of the metrological tool based on signals received from the ultrasonic transducer.

[0008] In at least one example, a user interface is in communication with the control unit. The user interface includes a display. The control unit is configured to show information regarding the ultrasonic transducer on the display.

[0009] In at least one example, the rod is formed of acrylic. Optionally, the rod can be formed of metal, or another homogenous material.

[0010] Each of the test holes includes a flat bottom. Two or more of the test holes can differ in depth. Two or more of the test holes can differ in diameter. Two or more of the test holes can be spaced apart from one another by an arc length.

[0011] In at least one example, the rod includes a first axial half and a second axial half. The test holes are formed in the first axial half. The second axial half is devoid of the test holes.

[0012] In at least one example, a diameter of the rod is two times a radius of a linear array of the ultrasonic transducer having a radius correction factor less than 1. In at least one example, a diameter of the rod is four times a radius of a linear array of the ultrasonic transducer having a radius correction factor of 1 or greater than 1.

[0013] The rod can include a pocket configured to receive and radially constrain an end of the ultrasonic transducer.

[0014] Certain examples of the present disclosure provide a method for assessing operational integrity of an ultrasonic transducer, the method including coupling the ultrasonic transducer to a metrological tool including a rod having test holes; and emitting, by the ultrasonic transducer, ultrasonic energy into the rod.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 illustrates a block diagram of a system, according to an example of the present disclosure.

[0016] FIG. 2 illustrates an isometric view of a metrological tool, according to an example of the present disclosure.

[0017] FIG. 3 illustrates a bottom view of the metrological tool of FIG. 2.

[0018] FIG. 4 illustrates an end view of the metrological tool of FIG. 2.

[0019] FIG. 5 illustrates a cross-sectional view of an ultrasonic transducer having a radius correction factor less than 1 disposed on a metrological tool through line 5-5 of FIG. 3, according to an example of the present disclosure.

[0020] FIG. 6 illustrates a cross-sectional view of an ultrasonic transducer having a radius correction factor of 1 or greater than 1 disposed on a metrological tool through line 5-5 of FIG. 3, according to an example of the present disclosure.

[0021] FIG. 7 illustrates a schematic block diagram of a control unit, according to an example of the present disclosure.

[0022] FIG. 8 illustrates a perspective front view of an aircraft, according to an example of the present disclosure.

[0023] FIG. 9 illustrates a perspective top view of ultrasonic transducers disposed on a structure, according to an example of the present disclosure.

[0024] FIG. 10 illustrates a radius correction factor as applied to an ultrasonic transducer and part.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0025] The foregoing summary, as well as the following detailed description of certain examples will be better understood when read in conjunction with the appended drawings. As used herein, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not necessarily excluding the plural of the elements or steps. Further, references to “one example” are not intended to be interpreted as excluding the existence of additional examples that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, examples “comprising” or “having” an element or a plurality of elements having a particular condition can include additional elements not having that condition.

[0026] FIG. 1 illustrates a block diagram of a system **100**, according to an example of the present disclosure. The system **100** includes a metrological tool **102** configured to assess operational integrity of an ultrasonic transducer **104**, which can be used to non-destructively inspect structural components. In at least one example, the ultrasonic transducer **104** includes a linear array **106** of ultrasonic elements **108** configured to emit ultrasonic energy.

[0027] The ultrasonic transducer **104** is in communication with a control unit **110**, such as through one or more wired or wireless connections. In at least one example, the control unit **110** is separate and distinct from the ultrasonic transducer **104**. For example, the control unit **110** can be part of a computer workstation, a handheld tool (such as a smart tablet or smart phone), or the like. As another example, the ultrasonic transducer **104** can include the control unit **110**.

[0028] The control unit **110** is in communication with a user interface **112**, such as through one or more wired or wireless connections. In at least one example, the user interface **112** includes a display **114** and an input tool **116**. In at least one example, the display **114** is an electronic device configured to electronically show images, videos, text, and/or the like. The display **114** can be a monitor, screen, television, touchscreen, and/or the like. The input tool **116** can include a keyboard, mouse, stylus, touchscreen interface (that is, the input tool **116** can be integral with the display **114**), and/or the like. The user interface **112** can be, or part of, a computer workstation, a handheld tool, or the like. The control unit **110** and the user interface **112** can be part of a common sub-

system, such as a computer workstation, a handheld tool, or the like. As another example, the user interface **112** can be separate and distinct from the control unit **110**.

[0029] The control unit **110** includes or is in communication with a memory that stores data regarding the metrological tool **102**. For example, the memory stores a sonogram detailing the structural features of the metrological tool **102**.

[0030] In operation, the ultrasonic transducer **104** is operated to emit ultrasonic energy into the metrological tool **102**. For example, the ultrasonic elements **108** of the linear array **106** emit ultrasonic energy into the metrological tool **102**. The ultrasonic energy reflects back from internal structural features of the metrological tool **102**. The control unit **110** receives signals regarding the emitted ultrasonic energy and reflected ultrasonic energy from the ultrasonic transducer **104** and forms a sonogram of the metrological tool **102** from the signals. The control unit **110** compares the resulting sonogram with the stored data regarding the metrological tool **102**. If the resulting sonogram conforms to (for example, is the same) as the stored data regarding the metrological tool **102**, the control unit **110** verifies the operational integrity of the ultrasonic transducer **104**. Conversely, if the resulting sonogram differs from the stored data, the control unit **110** determines that the ultrasonic transducer **104** has a fault.

[0031] In at least one example, the control unit **110** can show the resulting sonogram on the display **114** of the user interface **112**. The control unit **110** can show the resulting sonogram and the stored data of the metrological tool **102** (such as a specification sonogram) on the display **114**. The control unit **110** can also output signals to the user interface **112** regarding the determined assessment of the ultrasonic transducer **104** on the display **114**, such as text and/or a graphic indicating verification of the operational integrity of the ultrasonic transducer **104**, or an alert indicating a fault with the ultrasonic transducer **104**.

[0032] FIG. 2 illustrates an isometric view of the metrological tool **102**, according to an example of the present disclosure. The metrological tool **102** includes a main body **120** shaped as rod **122**, such as a cylindrical rod. In at least one example, the rod **122** includes a circular cross-section. The rod **122** extends between a first end **124** and a second end **126** opposite from the first end **124**. In at least one example, the rod **122** is formed of acrylic. In particular, rod **122** is a solid piece of acrylic. In at least one other example, the rod **122** is formed of a metal, or another homogenous material.

[0033] Test holes **130a**, **130b**, and **130c** are formed in the rod **122**. The test holes **130a**, **130b**, and **130c** are machined into the rod **122**. Each test hole **130a**, **130b**, and **130c** includes a flat bottom **132** disposed into the rod **122**. The flat bottom **132** connects to interior walls **134** that connect to an opening **136**.

[0034] The test holes **130a** have a first depth **138** that differs from a second depth **140** of the test holes **130b**, which, in turn, differs from a third depth **142** of the test holes **130c**. The first depth **138** is less than the second depth **140**, which is less than the third depth **142**. Each of the test holes **130a** can have the same first depth **138**. Optionally, depths of at least two of the test holes **130a** can be different. Similarly, each of the test holes **130b** can have the same second depth **140**. Optionally, depths of at least two of the test holes **130b** can be different. Also, similarly, each of the test holes **130c** can have the same third depth **142**. Optionally, depths of at least two of the test holes **130c** can be different.

[0035] The test holes **130a** have a diameter **144** that can be the same or different from a diameter **146** of the test holes **130b**, and the same or different from a diameter **148** of the test holes **130c**. Each of the test holes **130a** can have the same diameter **144**. Optionally, diameters of at least two of the test holes **130a** can differ. Similarly, each of the test holes **130b** can have the same diameter **146**. Optionally, diameters of at least two of the test holes **130b** can differ. Also, similarly, each of the test holes **130c** can have the same diameter **148**. Optionally, diameters of at least two of the test holes **130c** can differ.

[0036] The test holes **130a** have centers along an arc **160**, such as between 15-180 degrees. Each of the test holes **130a** is separated by a defined arc length. The test holes **130a** are at different arcuate

positions. The arc length between neighboring (that is, closest) test holes **130a** can be the same, or different. Similarly, the test holes **130b** have centers along an arc **162**, such as between 15-180 degrees. Each of the test holes **130b** is separated by a defined arc length. The test holes **130b** are at different arcuate positions. The arc length between neighboring (that is, closest) test holes **130b** can be the same, or different. Also, similarly, the test holes **130c** have centers along an arc **166**, such as between 15-180 degrees. Each of the test holes **130c** is separated by a defined arc length. The test holes **130c** are at different arcuate positions. The arc length between neighboring (that is, closest) test holes **130c** can be the same, or different.

[0037] In at least one example, the test holes **130a** are at a first axial position along a length of the rod **122**, while the test holes **130b** are at a second axial position along the length of the rod **122**, and the test holes **130c** are at a third axial position along the length of the rod **122**. The first, second, and third axial positions differ from one another. Optionally, one or more of the test holes **130a**, **130b**, and **130c** can be at a common axial position along the length of the rod.

[0038] As shown, the test holes **130a**, **130b**, and **130c** are formed in a portion of the rod **122**. In particular, the test holes **130a**, **130b**, and **130c** are formed within an arc length of 180 degrees or less. The test holes **130a**, **130b**, and **130c** are formed in a first portion **170**, such as a first axial half, of the rod **122**, but not a second portion **172**, such as a second axial half of the rod **122**. The second portion **172** is devoid of the test holes **130a**, **130b**, and **130c**. Referring to FIGS. **1** and **2**, during an assessment process, the ultrasonic transducer **104** is disposed on the second portion **172** so that ultrasonic energy is transmitted through the rod **122** and onto and/or into the test holes **130a-130c**, and ultrasonic energy is reflected back through the rod **122** without having any holes interfere with the reflected ultrasonic energy.

[0039] As shown, the metrological tool **102** includes three test holes **130a**, three test holes **130b**, and three test holes **130c**. Optionally, the metrological tool **102** can include more or fewer test holes **130a**, **130b**, and **130c** than shown. Further, the test holes **130a**, **130b**, and **130c** can extend over arcs that are greater or less than shown. Also, optionally, the metrological tool **102** can include only the test holes **130a**, **130b**, or **130c**. As another example, the metrological tool **102** can include the test holes **130a** and **130b**, but not the test holes **130c**. As another example, the metrological tool **102** can include the test holes **130b** and **130c**, but not the test holes **130a**. As another example, the metrological tool **102** can include the test holes **130a** and **130c**, but not the test holes **130b**. As another example, the metrological tool **102** can include test holes in addition to the test holes **130a-130c**.

[0040] The test holes **130a-130c** provide structural features within the metrological tool that can be mapped and stored as part of a specification sonogram stored within a memory. The control unit **110** compares the specification sonogram with a sonogram formed in response to the ultrasonic transducer **104** emitting the ultrasonic energy into the metrological tool **102**.

[0041] Referring to FIGS. **1** and **2**, the system **100** is for assessing operational integrity of the ultrasonic transducer **104**. The system **100** includes the metrological tool **102** including the rod **122** having a plurality of test holes **130a-c** formed in the rod **122**. The ultrasonic transducer **104** is configured to be coupled to (such as by abutting into) the metrological tool **102** and emit ultrasonic energy into the rod **122**. The ultrasonic energy is reflected off the test holes **130a-c**, which is then used to provide a sonogram of the metrological tool **102**. The sonogram can then be compared to structural data of the metrological tool **102** to assess the operational integrity of the ultrasonic transducer **104**. The control unit **110** is in communication with the ultrasonic transducer **104**, and is configured to generate the sonogram of the metrological tool **102** based on signals received from the ultrasonic transducer **104**.

[0042] FIG. **3** illustrates a bottom view of the metrological tool **102** of FIG. **2**. FIG. **4** illustrates an end view of the metrological tool of FIG. **2**. Referring to FIGS. **1-4**, in at least one example, the second portion **172** of the rod **122** is a bottom half. Optionally, the second portion **172** can be a top half.

[0043] As described herein, the metrological tool **102** is configured to verify operational integrity (such as array performance, depth of inspection, and coverage) of the ultrasonic transducer **104**, which is configured for non-destructive inspection. The metrological tool **102** includes the rod **122**, formed of acrylic. A series of flat-bottomed test holes **130a-c** are drilled at a known diameter, depth, and clocking. During an assessment process, the ultrasonic transducer **104** emits ultrasonic energy into the metrological tool **102**, and a reflected signal is analyzed by the control unit **110** in relation to stored data regarding the metrological tool **102** to determine the operational integrity of the ultrasonic transducer **104**.

[0044] In at least one example, the control unit **110** compares the reflected signal (for example, the reflected ultrasonic energy) to a theoretical signal (such as stored in memory) to verify performance of the ultrasonic transducer **104**. A diameter **190** of the rod **122** is selected according to a radius correction factor (RCF), and flat bottom test holes are drilled at depths to validate near and far surface detectability. The flat bottom test holes are clocked at various depths in order to prove array coverage of synthetic flaws (that is, the test holes. As shown in FIG. **10**, the RCF is determined as follows:

$$RCF=(r+d)/Rt$$

Where Rt is a transducer array radius, r is an inner surface radius of curvature of a part to be inspected by the ultrasonic transducer, and d is a maximum depth below an inner surface of the part. In at least one example, for ultrasonic transducers **104** having a an RCF less than 1, a diameter **190** of the rod **122** is two times a radius of the linear array **106**. In at least one example, for ultrasonic transducers **104** having an RCF greater than 1, the diameter **180** of the rod **122** is four times the radius of the linear array **106**.

[0045] Examples of the present disclosure provide systems and methods for validating the ultrasonic transducer **104** well before qualification, and do not require a production part for metrology testing. Further, the metrological tool **102** can also be used during non-destructive inspection system development, which can accelerate aperture selection and procedures. The metrological tool **102** can also lead to standardization of array performance, and thereby be used by several suppliers.

[0046] FIG. **5** illustrates a cross-sectional view of an ultrasonic transducer **104** having a radius correction factor less than 1 disposed on a metrological tool **102** through line 5-5 of FIG. **3**, according to an example of the present disclosure. As shown, the test holes **130a**, **130b**, and **130c** can have centers along a common cross-sectional arc length. The tests holes **130a**, **130b**, and **130c** have flat bottoms **132** at different depths. The example shown in FIG. **5** can be used with respect to an RCF of 1 or less.

[0047] FIG. **6** illustrates a cross-sectional view of an ultrasonic transducer **103** having a radius correction factor of 1 or greater than 1 disposed on a metrological tool **102** through line 5-5 of FIG. **3**, according to an example of the present disclosure. In this example, a pocket **194** can be formed in the rod **122**. The pocket **194** can extend along a length of the rod **122**. The pocket **194** conforms to a size and a shape of an end **196** of the ultrasonic transducer **104**. The end **196** includes the linear array **106** of ultrasonic elements **108** (shown in FIG. **1**). The pocket **194** is configured to receive and radially constrain the end **196** to allow for a centerline of the linear array **106** to match a centerline **198** of the rod **122**. The example shown in FIG. **6** can be used with respect to an RCF between 1 and 2.

[0048] FIG. **7** illustrates a schematic block diagram of the control unit **110**, according to an example of the present disclosure. In at least one example, the control unit **110** includes at least one processor **200** in communication with a memory **202**. The memory **202** stores instructions **204**, received data **206**, and generated data **208**. The control unit **110** shown in FIG. **7** is merely exemplary, and non-limiting.

[0049] As used herein, the term “control unit,” “central processing unit,” “CPU,” “computer,” or

the like may include any processor-based or microprocessor-based system including systems using microcontrollers, reduced instruction set computers (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor including hardware, software, or a combination thereof capable of executing the functions described herein. Such are exemplary only, and are thus not intended to limit in any way the definition and/or meaning of such terms. For example, the control unit **110** may be or include one or more processors that are configured to control operation, as described herein.

[0050] The control unit **110** is configured to execute a set of instructions that are stored in one or more data storage units or elements (such as one or more memories), in order to process data. For example, the control unit **110** may include or be coupled to one or more memories. The data storage units may also store data or other information as desired or needed. The data storage units may be in the form of an information source or a physical memory element within a processing machine.

[0051] The set of instructions may include various commands that instruct the control unit **110** as a processing machine to perform specific operations such as the methods and processes of the various examples of the subject matter described herein. The set of instructions may be in the form of a software program. The software may be in various forms such as system software or application software. Further, the software may be in the form of a collection of separate programs, a program subset within a larger program, or a portion of a program. The software may also include modular programming in the form of object-oriented programming. The processing of input data by the processing machine may be in response to user commands, or in response to results of previous processing, or in response to a request made by another processing machine.

[0052] The diagrams of examples herein may illustrate one or more control or processing units, such as the control unit **110**. It is to be understood that the processing or control units may represent circuits, circuitry, or portions thereof that may be implemented as hardware with associated instructions (e.g., software stored on a tangible and non-transitory computer readable storage medium, such as a computer hard drive, ROM, RAM, or the like) that perform the operations described herein. The hardware may include state machine circuitry hardwired to perform the functions described herein. Optionally, the hardware may include electronic circuits that include and/or are connected to one or more logic-based tools, such as microprocessors, processors, controllers, or the like. Optionally, the control unit **110** may represent processing circuitry such as one or more of a field programmable gate array (FPGA), application specific integrated circuit (ASIC), microprocessor(s), and/or the like. The circuits in various examples may be configured to execute one or more algorithms to perform functions described herein. The one or more algorithms may include aspects of examples disclosed herein, whether or not expressly identified in a flowchart or a method.

[0053] As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in a data storage unit (for example, one or more memories) for execution by a computer, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above data storage unit types are exemplary only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0054] FIG. **8** illustrates a perspective front view of an aircraft **300**, according to an example of the present disclosure. The aircraft **300** includes a propulsion system **312** that includes engines **314**, for example. Optionally, the propulsion system **312** may include more engines **314** than shown. The engines **314** are carried by wings **316** of the aircraft **300**. In other examples, the engines **314** may be carried by a fuselage **318** and/or an empennage **320**. The empennage **320** may also support horizontal stabilizers **322** and a vertical stabilizer **324**. The fuselage **318** of the aircraft **300** defines an internal cabin **330**, which includes a flight deck or cockpit, one or more work sections (for example, galleys, personnel carry-on baggage areas, and the like), one or more passenger sections (for example, first class, business class, and coach sections), one or more lavatories, and/or the like.

FIG. 8 shows an example of an aircraft 300. It is to be understood that the aircraft 300 can be sized, shaped, and configured differently than shown in FIG. 8.

[0055] FIG. 9 illustrates a perspective top view of ultrasonic transducers 104 disposed on a structure 400, according to an example of the present disclosure. The structure 400 can be a portion of an aircraft, such as the aircraft 300 shown in FIG. 8. For example, the structure 400 can be a stringer. The ultrasonic transducers 104 are coupled to the structure 400 and operated to perform non-destructive inspection of the structure 400. Before the ultrasonic transducers 104 are coupled to the structure 400, the ultrasonic transducers 104 can be assessed by the system 100 including the metrological tool 102, as shown and described with respect to FIGS. 1-7.

[0056] Further, the disclosure comprises examples according to the following clauses:

[0057] Clause 1. A system for assessing operational integrity of an ultrasonic transducer, the system comprising: [0058] a metrological tool including a rod having test holes, wherein the ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the rod.

[0059] Clause 2. The system of Clause 1, further comprising a control unit in communication with the ultrasonic transducer, wherein the control unit is configured to generate a sonogram of the metrological tool based on signals received from the ultrasonic transducer.

[0060] Clause 3. The system of Clause 2, further comprising a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the ultrasonic transducer on the display.

[0061] Clause 4. The system of any of Clauses 1-3, wherein the rod is formed of acrylic.

[0062] Clause 5. The system of any of Clauses 1-4, wherein each of the test holes includes a flat bottom.

[0063] Clause 6. The system of any of Clauses 1-5, wherein two or more of the test holes differ in depth.

[0064] Clause 7. The system of any of Clauses 1-6, wherein two or more of the test holes differ in diameter.

[0065] Clause 8. The system of any of Clauses 1-7, wherein two or more of the test holes are spaced apart from one another by an arc length.

[0066] Clause 9. The system of any of Clauses 1-8, wherein the rod comprises a first axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is devoid of the test holes.

[0067] Clause 10. The system of any of Clauses 1-9, wherein a diameter of the rod is two times a radius of a linear array of the ultrasonic transducer having a radius correction factor less than 1.

[0068] Clause 11. The system of any of Clauses 1-10, wherein a diameter of the rod is four times a radius of a linear array of the ultrasonic transducer having a radius correction factor greater than 1.

[0069] Clause 12. The system of any of Clauses 1-11, wherein the rod comprises a pocket configured to receive and radially constrain an end of the ultrasonic transducer.

[0070] Clause 13. A method for assessing operational integrity of an ultrasonic transducer, the method comprising: [0071] coupling the ultrasonic transducer to a metrological tool including a rod having test holes; and [0072] emitting, by the ultrasonic transducer, ultrasonic energy into the rod.

[0073] Clause 14. The method of Clause 13, further comprising: [0074] generating, by a control unit in communication with the ultrasonic transducer, a sonogram of the metrological tool based on signals received from the ultrasonic transducer; and [0075] showing, by the control unit on a display of a user interface, information regarding the ultrasonic transducer on the display.

[0076] Clause 15. The method of Clauses 13 or 14, wherein the rod is formed of acrylic, and wherein each of the test holes includes a flat bottom.

[0077] Clause 16. The method of any of Clauses 13-15, wherein two or more of the test holes differ in depth, wherein two or more of the test holes differ in diameter, and wherein two or more of the test holes are spaced apart from one another by an arc length, wherein the rod comprises a first

axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is devoid of the test holes.

[0078] Clause 17. The method of any of Clauses 13-16, wherein a diameter of the rod is two times a radius of a linear array of the ultrasonic transducer having a radius correction factor less than 1.

[0079] Clause 18. The method of any of Clauses 13-17, wherein a diameter of the rod is four times a radius of a linear array of the ultrasonic transducer having a radius correction factor greater than 1.

[0080] Clause 19. The method of any of Clauses 13-18, further comprising receiving and radially constraining an end of the ultrasonic transducer within a pocket formed in the rod.

[0081] Clause 20. A system for assessing operational integrity of an ultrasonic transducer, the system comprising: [0082] a metrological tool including an acrylic rod having test holes, wherein the ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the acrylic rod, wherein each of the test holes includes a flat bottom, wherein two or more of the test holes differ in depth, wherein two or more of the test holes differ in diameter, and wherein two or more of the test holes are spaced apart from one another by an arc length, and wherein the acrylic rod comprises a first axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is devoid of the test holes; [0083] a control unit in communication with the ultrasonic transducer, wherein the control unit is configured to generate a sonogram of the metrological tool based on signals received from the ultrasonic transducer; and [0084] a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the ultrasonic transducer on the display.

[0085] As described herein, examples of the present disclosure provide systems and methods for verifying operational integrity of an ultrasonic transducer, such as used for non-destructive inspection of structural components.

[0086] While various spatial and directional terms, such as top, bottom, lower, mid, lateral, horizontal, vertical, front and the like can be used to describe examples of the present disclosure, it is understood that such terms are merely used with respect to the orientations shown in the drawings. The orientations can be inverted, rotated, or otherwise changed, such that an upper portion is a lower portion, and vice versa, horizontal becomes vertical, and the like.

[0087] As used herein, a structure, limitation, or element that is “configured to” perform a task or operation is particularly structurally formed, constructed, or adapted in a manner corresponding to the task or operation. For purposes of clarity and the avoidance of doubt, an object that is merely capable of being modified to perform the task or operation is not “configured to” perform the task or operation as used herein.

[0088] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described examples (and/or aspects thereof) can be used in combination with each other. In addition, many modifications can be made to adapt a particular situation or material to the teachings of the various examples of the disclosure without departing from their scope. While the dimensions and types of materials described herein are intended to define the aspects of the various examples of the disclosure, the examples are by no means limiting and are exemplary examples. Many other examples will be apparent to those of skill in the art upon reviewing the above description. The scope of the various examples of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims and the detailed description herein, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112 (f), unless and until such

claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

[0089] This written description uses examples to disclose the various examples of the disclosure, including the best mode, and also to enable any person skilled in the art to practice the various examples of the disclosure, including making and using any tools or systems and performing any incorporated methods. The patentable scope of the various examples of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if the examples have structural elements that do not differ from the literal language of the claims, or if the examples include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A system for assessing operational integrity of an ultrasonic transducer, the system comprising: a metrological tool including a rod having test holes, wherein the ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the rod.
2. The system of claim 1, further comprising a control unit in communication with the ultrasonic transducer, wherein the control unit is configured to generate a sonogram of the metrological tool based on signals received from the ultrasonic transducer.
3. The system of claim 2, further comprising a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the ultrasonic transducer on the display.
4. The system of claim 1, wherein the rod is formed of acrylic.
5. The system of claim 1, wherein each of the test holes includes a flat bottom.
6. The system of claim 1, wherein two or more of the test holes differ in depth.
7. The system of claim 1, wherein two or more of the test holes differ in diameter.
8. The system of claim 1, wherein two or more of the test holes are spaced apart from one another by an arc length.
9. The system of claim 1, wherein the rod comprises a first axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is devoid of the test holes.
10. The system of claim 1, wherein a diameter of the rod is two times a radius of a linear array of the ultrasonic transducer having a radius correction factor less than 1.
11. The system of claim 1, wherein a diameter of the rod is four times a radius of a linear array of the ultrasonic transducer having a radius correction factor greater than 1.
12. The system of claim 1, wherein the rod comprises a pocket configured to receive and radially constrain an end of the ultrasonic transducer.
13. A method for assessing operational integrity of an ultrasonic transducer, the method comprising: coupling the ultrasonic transducer to a metrological tool including a rod having test holes; and emitting, by the ultrasonic transducer, ultrasonic energy into the rod.
14. The method of claim 13, further comprising: generating, by a control unit in communication with the ultrasonic transducer, a sonogram of the metrological tool based on signals received from the ultrasonic transducer; and showing, by the control unit on a display of a user interface, information regarding the ultrasonic transducer on the display.
15. The method of claim 13, wherein the rod is formed of acrylic, and wherein each of the test holes includes a flat bottom.
16. The method of claim 13, wherein two or more of the test holes differ in depth, wherein two or more of the test holes differ in diameter, and wherein two or more of the test holes are spaced apart from one another by an arc length, wherein the rod comprises a first axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is

devoid of the test holes.

17. The method of claim 13, wherein a diameter of the rod is two times a radius of a linear array of the ultrasonic transducer having a radius correction factor less than 1.

18. The method of claim 13, wherein a diameter of the rod is four times a radius of a linear array of the ultrasonic transducer having a radius correction factor greater than 1.

19. The method of claim 13, further comprising receiving and radially constraining an end of the ultrasonic transducer within a pocket formed in the rod.

20. A system for assessing operational integrity of an ultrasonic transducer, the system comprising: a metrological tool including an acrylic rod having test holes, wherein the ultrasonic transducer is configured to be coupled to the metrological tool and emit ultrasonic energy into the acrylic rod, wherein each of the test holes includes a flat bottom, wherein two or more of the test holes differ in depth, wherein two or more of the test holes differ in diameter, and wherein two or more of the test holes are spaced apart from one another by an arc length, and wherein the acrylic rod comprises a first axial half and a second axial half, wherein the test holes are formed in the first axial half, and wherein the second axial half is devoid of the test holes; a control unit in communication with the ultrasonic transducer, wherein the control unit is configured to generate a sonogram of the metrological tool based on signals received from the ultrasonic transducer; and a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the ultrasonic transducer on the display.
